BAYESIAN NETWORK

CODE:

```
!pip install pgmpy
from pgmpy.models import DiscreteBayesianNetwork
from pgmpy.estimators import ParameterEstimator
from pgmpy.inference import VariableElimination
import pandas as pd

# Sample data
data = pd.DataFrame(data={'Rain': ['No', 'No', 'Yes', 'Yes', 'No', 'Yes', 'Yes', 'No'],
            'TrafficJam': ['Yes', 'No', 'Yes', 'No', 'Yes', 'Yes', 'No', 'No'],
            'ArriveLate': ['Yes', 'No', 'Yes', 'No', 'No', 'Yes', 'Yes', 'No']})

# Define the Bayesian network structure
model = DiscreteBayesianNetwork([('Rain', 'TrafficJam'), ('TrafficJam', 'ArriveLate')])

# Fit the model to the data using Maximum Likelihood Estimation
model.fit(data)

# Print conditional probability distributions
print(model.get_cpds())

# Perform inference
inference = VariableElimination(model)
query_result = inference.query(variables=['ArriveLate'], evidence={'Rain': 'Yes'})
print(query_result)
```

OUTPUT:

```
[<TabularCPD representing P(Rain:2) at 0x7eb632a27f20>,
+-----------------+-------------------+
| ArriveLate      |   phi(ArriveLate) |
+=================+===================+
| ArriveLate(No)  |            0.5000 |
+-----------------+-------------------+
| ArriveLate(Yes) |            0.5000 |
+-----------------+-------------------+
```

CASE STUDY:Loan Default Prediction

```
!pip install pgmpy

from pgmpy.models import DiscreteBayesianNetwork
from pgmpy.factors.discrete import TabularCPD
from pgmpy.inference import VariableElimination

model = DiscreteBayesianNetwork([('Income', 'Default'), ('Credit', 'Default')])

cpd_income = TabularCPD('Income', 2, [[0.6], [0.4]])
cpd_credit = TabularCPD('Credit', 2, [[0.7], [0.3]])
cpd_default = TabularCPD('Default', 2,
                    [[0.95, 0.8, 0.9, 0.5],
                     [0.05, 0.2, 0.1, 0.5]],
                    evidence=['Income', 'Credit'], evidence_card=[2, 2])

model.add_cpds(cpd_income, cpd_credit, cpd_default)

infer = VariableElimination(model)
query_result = infer.query(['Default'], evidence={'Income': 1, 'Credit': 0})
print(query_result)
```

OUTPUT:

```
+------------+----------------+
| Default    |   phi(Default) |
+============+================+
| Default(0) |         0.9000 |
+------------+----------------+
| Default(1) |         0.1000 |
+------------+----------------+
```

CASE STUDY: DISEASE DETECTION

```
!pip install pgmpy

from pgmpy.models import DiscreteBayesianNetwork
from pgmpy.factors.discrete import TabularCPD
from pgmpy.inference import VariableElimination

model = DiscreteBayesianNetwork([('Smoking', 'Cancer'), ('Pollution', 'Cancer')])

cpd_smoking = TabularCPD('Smoking', 2, [[0.7], [0.3]])
cpd_pollution = TabularCPD('Pollution', 2, [[0.6], [0.4]])
cpd_cancer = TabularCPD('Cancer', 2,
                    [[0.99, 0.85, 0.90, 0.60],
                     [0.01, 0.15, 0.10, 0.40]],
                    evidence=['Smoking', 'Pollution'], evidence_card=[2, 2])

model.add_cpds(cpd_smoking, cpd_pollution, cpd_cancer)

infer = VariableElimination(model)
query_result = infer.query(['Cancer'], evidence={'Smoking': 1, 'Pollution': 1})
print(query_result)
```

OUTPUT:

```
+-----------+---------------+
| Cancer    |   phi(Cancer) |
+===========+===============+
| Cancer(0) |        0.6000 |
+-----------+---------------+
| Cancer(1) |        0.4000 |
+-----------+---------------+
```